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REGULAR ARTICLES

Development and evaluation of a herbal anti-tick grease using *Cissus quadrangularis* terpenes and terpenoids as active compounds for controlling ticks on live stock

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Abstract
Ticks and tick borne diseases are on the rise. This is leading to diseases of economic importance in animals such that farmer`s budgetary allocation is constrained. Recently, interventions involving the use of plant based anti-tick products has shown promising results. Hence, in the present study, a tick grease was developed and evaluated utilising terpenes and terpenoids from *Cissus quadrangularis* as principal active components. Terpene and terpenoid rich extracts were isolated using a normal phase silica gel packed column and active fraction determined using repellence and knockdown effect. Ethyl acetate was used to extract the terpenes and terpenoids from dried *Cissus quadrangularis* stems. The chemical composition of the active fraction was evaluated using GC–MS. The formulated tick grease was evaluated under both laboratory and field studies. Out of the six column isolated fractions only 2, eluted with hexane: ethyl acetate ratio: 8:2 (100 mL), and 7:3 (100 mL), showed significant acaricidal activity with a percentage knockdown effect of 100 and 75% respectively. The GC–MS analysis of the combine two fractions showed that, they consisted of various monoterpenes and sesquiterpenes. Copaene (14.56%), α-ocimene (42.23%), murolene (16.98%) and caryophyllene oxide (11.89%) were found to be the major occurring compounds in the fraction. The tick grease formulated with the monoterpenes and sesquiterpenes rich fraction showed significant acaricidal activity when compared to commercial acaricide, armitraz purchased from local veterinary shops. The tick grease showed percent repellence of 100% as compared to 75% for the commercial acaricide. Under field studies, by topical application on cows the tick grease knocked down all ticks including *Rhipicephalus appendiculatus* and *Amblyomma variegatum* species which were resistant to the commercial acaricide. It also showed good protective effect over the 5 day period selected for the study on non-tick infested cows. Thus our study prove the potential of using terpenes and terpenoids from *Cissus quadrangularis* as active compounds in formulating a tick grease.

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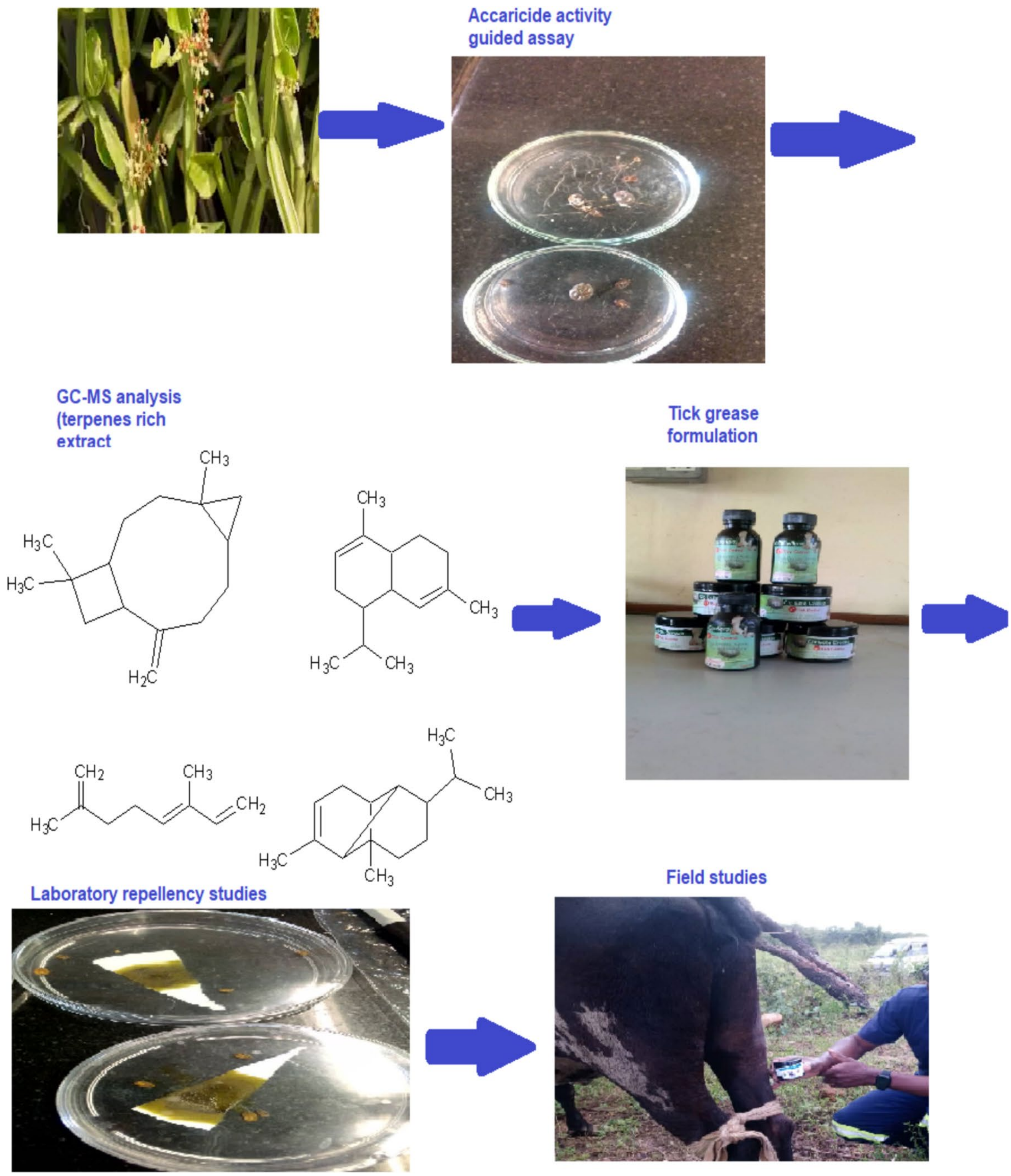
Graphical Abstract



Keywords *Cissus quadrangularis* · Herbal acaricides · Terpenes · Terpenoids · Tick grease

Introduction

Globally productivity in livestock farming continue to declined due to high death rates caused by tick borne diseases such as *Theileriosis* and heart water (Upadhaya et al. 2020; Fouche et al. 2016). The silent pandemics have caused chaos among both small scale and large scale farmers (Kumar et al. 2019). Emergence of serious resistance to current conventional acaricides is furthermore complicating the problem (Vimonish et al. 2021). Ticks, classified as haematophagous ectoparasites, targets multiple hosts and portray a wide geographic diversity (Gupta et al. 2018). When they bite an animal they infect it with a protozoan, bacterial, rickettsial and viral diseases which are the major diseases that are severely fatal to both livestock, humans and domestic animals (Asri et al. 2023). Tick mediated protozoan diseases such as Theileriosis, babesiosis, rickettsial and cowdriosis are the most common diseases affecting the livelihoods of farming communities in developing countries and the global world (Nyahangare et al. 2015). Apart from transmitting diseases, tick infestations affects negatively live weight and milk production (Rajput et al. 2006). Due to severity effects of ticks, a substantial proportion of the annual input costs of many cattle farmers now go to management and control of ticks and tick-borne diseases (Muvhuringi et al. 2022). Approximately economic losses of US$720 million, US$100 million and US$1 billion annualy are realized in Africa, Australia and South America respectively (Horn 1987; Cobon and Willadsen 1990; Minjauw and McLeod 2003). Losses from the cost of acaricides, milk losses and treatment costs were estimated at US$6 million per annum in Zimbabwe over a 10-year period (Adenubi 2017). Based on this information, it is evident that ticks and the diseases they transmit have become a major drawback to the improvement of the livestock industry (Lorusso et al. 2016). Current control methods are largely based on the use of commercially available chemicals such as the arsenicals, chlorinated hydrocarbons, organophosphates, carbamates, formamidines, pyrethroids, macrocyclic lactones, and more recently insect growth regulators (Bisht et al. 2021). These products face serious challenges of development of resistance and have been implicated in having environmentally destructive tendencies (Kumar et al. 2017; Wanzala 2017). Persistence and environmental contamination due to continued use of such chemically based acaricides have been reported, leading to restrictions on their use thus limiting the available options for controlling tick borne diseases (Kumar et al. 2023). Scientific studies now should seek to produce highly efficacious and environmentally friendly products to control ticks and tick borne diseases so as to prevent the demise of livestock farming.

The novel use of ethnobotanicals to pest and diseases management and control is a potential solution to the primary health care of animals (Nyahangare et al. 2015). This is because, they are believed to be safe (Muraleedharan 2019) and consist of many phytochemicals that can help arrest resistance (Sanhokwe et al. 2015). Previous studies have shown significant in vitro potency of plant extracts in controlling ticks (Shanmuganath et al. 2021). However, the knowledge on the use of them is passed orally and due to socio-economic changes the information might disappear (Sanhokwe et al. 2015). Thus, there is need for documentation of such information for use in the next generations were possible performing formulation studies so that they can be used in cases of serious tick resistance.

In the present study, terpenes and terpenoids from *Cissus quadrangularis* were used as active principal components to formulate a herbal tick grease. The *C. quadrangularis* aqueous plant extracts has been extensively used by farmers in Southern Africa for the control of ticks due to its bio-insecticidal properties (Hamid and Patil 2023). It is a plant native to India and Africa, (Doss 2015). It is documented in the Ayuverdic medicine for its effectiveness in bone fracture healing, (Mukherjee et al. 2016). Other uses include anti-ulcer activity, anti-inflammatory activity, wound healing properties, weight loss and tissue repair, (Stohs and Ray 2013). *Cissus quadrangularis* is a plant that has been found to be rich in terpenes and terpenoids and can be propagated easily, therefore making it a good candidate for bio prospecting (Kaur et al. 2022). Terpenes and terpenoids are phytocompounds that have shown significant acaricidal activity (Cardosoa et al. 2020). In this study we investigated the in vitro and in vivo acaricidal activity of a topical grease formulated using *Cissus quadrangularis* terpenes and terpenoids against *Rhipicephalus appendiculatus, Amblyomma variegatum* and *Hyalomma truncatum* tick species.

Materials and methods

Plant sample

Fresh stems, 2 kgs of the *Cissus quadrangularis* plant were collected from Macheke in February, 2023. The plant materials were further identified at the National Herbarium in Harare, Zimbabwe and voucher specimens 2023/1 CQ were deposited at Bindura University of Science Education natural product section for future reference. About 500 g were shade dried for 4 weeks and made into powder 0.75 microns using an electric grinder and sieve. These were stored in closed khaki envelopes until required. The remaining stems were further propagated at the university herbal garden for future use.

Tick collection

Ticks, *Rhipicephalus appendiculastus, Amblyomma variegatum* and *Hyalomma truncatum* were collected from different areas, Macheke, Sanyati, Mt Darwin, Makuti and Manhenga for the laboratory bioassays. Tick species identification was conducted by one of the researchers, Mr Mangwiro an animal scientist during collection. Collection from different areas were to cater for the inherent resistance issues. Both male and female engorged ticks were picked from the animals selected randomly from the cattle kraal. The presence of ticks especially adults which were engorged was detected on the hair coat of the host with the palm of hand as described by Latif and Walker (2004). To remove ticks, a strong steel forceps of medium size with blunt ends and serrated inner surfaces was used. The forceps was used to grip the tick firmly over its scutum and mouthparts as closely to the host's skin as possible, then the tick was pulled strongly and directly out as described by Latif and Walker (2004). The ticks were further identified by a Veterinary doctor and were classified as *Rhipicephalus appendiculatus, Amblyomma variegatum* and *Hyalomma truncatum* species.

Preparation of plant extracts

Shade dried powdered sample (100 g) of *Cissus quadrangularis* plant was macerated for 48 h in an absolute ethyl acetate solvent. Ethyl acetate was used because it had shown greater acaricide activity than hexane and methanol extract in a preliminary study results not shown. The contents were then filtered by Whatman number 1 filter paper under room temperature. The extracts were concentrated under vacuum to yield a gold coloured oily liquid.

Isolation of active terpenes and terpenoids

Exactly 1 ml of oily extract was dispersed in 1 ml ethyl acetate. A normal phase silica gel packed column (500 g) was then used to isolate the terpenes and terpenoids. The column was packed using hexane and the sample was carefully loaded on the column to form a band. Elution was done each time with increasing polarity (v/v) ratio of hexane: ethyl acetate 10:0 (100 mL), 9:1 (100 mL), 8:2 (100 mL), 7:3 (100 mL), 1:1 (100 mL) and final 100% ethyl acetate (100 mL). A total of 6 fractions were obtained which were concentrated under vacuum to almost dryness and dissolved in ethyl acetate.

Determination of bioactive fractions

This was assayed using contact toxicity using each fraction prepared in ethyl acetate. 1 ml of each of the solution was placed in sterile petri dish in triplicate and allowed to dry. Adult ticks (4) were placed into each petri dish. A different dropper was used to draw each solution as well as ethyl acetate (negative control) and the drops were placed on the live ticks to mimic manual dipping when using knapsack sprayers. Amitraz was used as a positive control. Observations were done on the percentage knock down of the ticks. Parameters used in evaluating the ticks as knocked down or alive included appendage movement, signs of cuticle darkness, haemorrhagic skin lesions and stopped Malpighian tube movement. Percentage knockdown was calculated using equation

$$\text{Percentage knockdown} = \frac{\textit{knocked down ticks} - \textit{live ticks}}{\textit{Total ticks}} \times 100$$

Composition analysis by GC–MS

Out of the six fractions only 2 showed acaricidal activity, F3 and F4. The 2 fractions were combined and analysed using Gas chromatography-Mass spectrometry (GC–MS) Thermo Fisher Scientific's (Waltham, MA, USA) TSQ 8000 Evo system, with auto sampler and TG-1MS capillary column. Helium was the carrier gas. Split ratio of 1:200 was used to introduce the sample. To evaluate the spectral data, Xcalibur 1.1 software was used. Identification of compounds was done by matching with the compounds in the NIST library. The temperature was gradually raised while the spectra were scanned. The relative percentages were ascertained by computation of peak areas of the constituent compounds.

Formulation of the tick grease

The combined fraction was diluted with ethyl acetate to varying concentrations 5, 10. 50 and 100% v/v. The composition of the tick grease was utilised as shown in the Table 1 below;

Table 1 Formulation of anti-tick grease

Ingredients	Quantities (%)				
	Fm1	Fm2	Fm3	Fm4	Fm5
Terpenes and terpenoids extract (F1 &F2)	5	10	50	100	0
Paraffin wax	10	10	10	10	10
Petroleum jelly	55	55	55	55	60
White mineral oil	29	29	29	29	29
MPG	1	1	1	1	1

Fm = formulation, *MPG* = Mono Propylene Glycol (F1 & F2) fraction 1 and 2 combined

The dissolved extract was first dispersed using a hand held stick blender in MPG and contents set aside, Phase 1. Wax and petroleum jelly was melted in a water bath followed by adding paraffin wax. The melted contents were removed and temperature allowed to drop to 45 oC and then Phase 1 was added and stirred to mix. The contents were packed into 100 ml black opaque containers and allowed to set.

Acaricidal activity of the formulated tick grease

The acaricidal activity of the tick grease was tested both under laboratory and field studies.

Laboratory studies

Contact and repellence studies were conducted in the laboratory to investigate the acaricidal activity of the tick grease. To investigate the repellent activity a filter paper was cut into a triangular shape and the grease applied at the middle, Fig. 2. The filter paper was then placed in a petri dish and five ticks were introduced so that they are not in contact with the treated area. The petri dishes were sealed immediately with a perforated lid to avoid fumigation and the number of ticks within the vicinity of the treated and untreated area were monitored per hour for 6 h. The percent repellence was calculated as follows:

$$\%\ repellence = \frac{[(Nu - Nt)]}{[(Nu + Nt)]} \times 100$$

where Nu is the number of ticks in the untreated area; Nt is the number of ticks in the treatment area.

Contact toxicity assays were conducted by applying the grease in the whole filter paper disc and then placed in petri dishes. Adult ticks (10) were placed into each petri dish so that they directly get in contact with the treated filter papers. The experiments were replicated 5 times. A commercial tick grease was used as a positive control. Observations were done on the percentage knock down of the ticks after 5 h. Parameters used in evaluating the ticks as knocked down or alive included appendage movement, signs of cuticle darkness, haemorrhagic skin lesions and stopped Malpighian tube movement. Percentage knockdown was calculated using equation.

$$\text{Percentage knockdown} = \frac{knocked\ down\ ticks - live\ ticks}{live\ ticks} \times 100$$

Field studies

Field studies were conducted in 3 stages, (i) without stimulus (ii) with stimulus (iii) and on cows. Formulation 3 showed the best repellence and toxicity effect under laboratory studies therefore it was selected for field studies. Initially tick grease was applied on a piece of cloth (an adult person new shirt) and was tied to a string. The treated shirt was dragged in a grass land which was infested with ticks. The results were compared with results obtained using a non-treated shirt. The assay with stimulus was conducted by the researchers by wrapping one leg with treated piece of cloth and leaving the other one without. The researchers then walked in the grassland for 20 min and after the number of ticks on the two legs were compared. The tick grease was then directly applied on infected and non-infested cows and the number of knock down and dead ticks were observed after 2 h. Untreated areas on the same cow were used as control. Number of new infestations were monitored every day for 5 days. Field experiments were conducted in January were the number of tick infestation was high, characterized by warm temperatures and high humidity conditions. Results were compared with results of a commercial tick grease. The safety of the tick grease was also checked by monitoring the behaviour of the cows for itchiness and on the applied area for irritation over the 5 day period.

Results

Determination of bioactive fractions

Table 2 show results of the determination of bioactive fractions isolated using a normal phase silica gel packed column (500 g). Fraction 3 (F3 and F4) showed significant knockdown activity. In comparison to the commercial acaricides F3 showed greater activity with 100% knockdown effect Fig. 1.

Because F3 and F4 exhibited significant knockdown effect they were combined and terpene composition analysed using Gas chromatography-Mass spectrometry (GC–MS). Table 3 shows terpenes and terpenoids

Table 2 Bioactivity of fractions isolated by silica gel packed column

Identity	F1	F2	F3*	F4*	F5	F6	Amitraz
Percentage knockdown	-	-	100 ± 0	75 ± 0	-	-	75 ± 0

Hexane: ethyl acetate ratio: F1, 10:0 (100 mL), F2, 9:1 (100 mL), F3, 8:2 (100 mL), F4, 7:3 (100 mL), F5, 1:1 (100 mL) and F6, 100% ethyl acetate (100 mL). *Not significantly different from Amitraz*

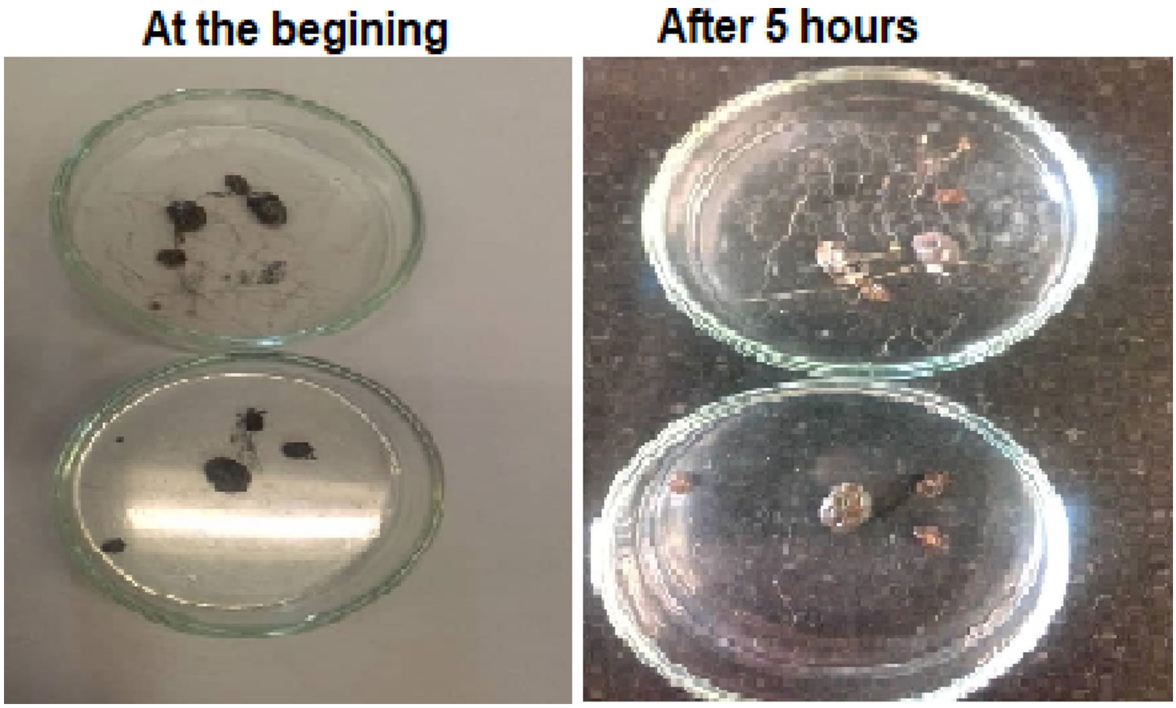


Fig. 1 Bioactive properties of fraction 3

Table 3 Terpenes and terpenoids GC–MS analysis of fraction 3 and 4

Peak number	Retention time	Name of compound	Relative similarity index	% Relative peak area
1	6.22	α-pinene	98.6	0.21
2	6.33	Limonene	94.5	0.27
3	7.81	α-ocimene	98.4	43.23
4	7.90	γ-terpinene	80.9	0.32
5	8.11	Carene	76.4	0.56
6	8.43	Terpenin-4-ol	95.6	0.26
7	8.79	Copaene	99.3	14.56
8	10.44	Isocaryophyllene	96.8	1.95
9	10.65	Caryophyllene	78.1	3.10
10	11.01	Murolene	98.9	16.98
11	12.43	Isoledene	75.5	3.67
12	12.87	Lilac alcohol D	87.0	0.66
13	12.91	Benzene, 1-(1,5-dimethylhexyl)−4-methyl-	89.7	1.65
14	13.40	Caryophyllene oxide	99.2	11.89
15	13.51	Aromadendrene epoxide	98.7	0.69
Total				100

Acaricidal activity of the formulated tick grease under laboratory conditions

composition of F3 and F4 composite sample. The sample consist of several compounds of which α-ocimene, copaene, murolene and caryophyllene oxide were in significant amounts basing on peak area.

Table 4 and Fig. 2 show results of repellence activity of formulations 1–5 consisting of different concentration of the terpene extracts. Formulation 3 and 4 consisting of 50 and 100% terepenes showed the greatest repellence activity.

Table 4 Percentage repellence of tick grease and standard commercial tick grease

	Percent repellence at					
	1 h	2h	3h	4h	5h	6 h
Fm 1	-	-	-	-	-	-
Fm 2	-	-	-	-	-	-
Fm 3	100	100	100	100	100	100
Fm 4	100	100	100	100	100	100
Fm 5	-	-	-	-	-	-
S	50	50	60	60	70	75

S = standard commercial tick grease

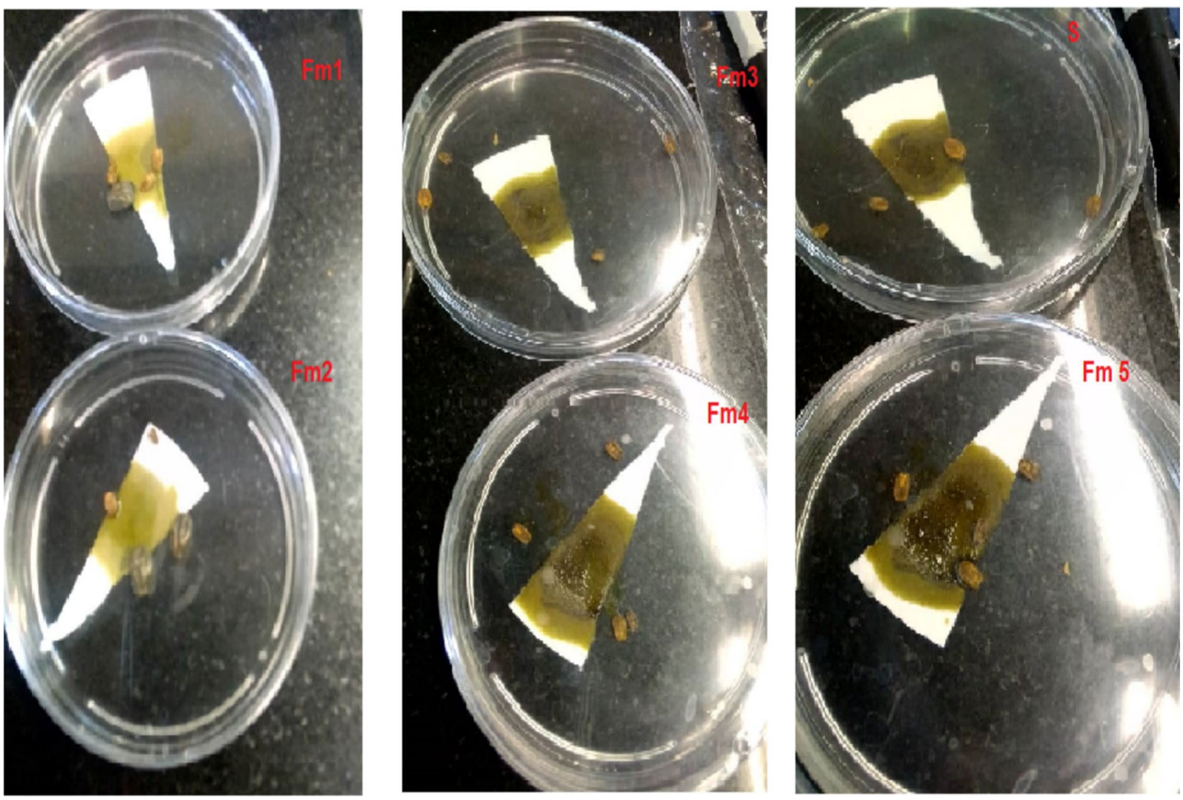


Fig. 2 Repellence activity of the tick grease, formulation 1–5 and standard S, Standard was a commercial tick grease. Fm 5 consisted of no terpenes extract

Formulation 5 consisting of no extract showed no repellence activity showing that the terpenes and terpenoids in the extract are responsible for the activity.

Toxicity of the tick grease

Table 5 presents toxicity of of the formulated grease under laboratory conditions. Formulation 3 and 4 at dosage of 50 and 100% exhibited greater percentage knocked down effect than commercial acaricides.

Field studies

The formulated herbal tick grease was tested to check its effectiveness under real environment conditions. Tables 6 and 7 shows results for assays without stimulus for treated and untreated cloth. The results for the treated cloth showed that the herbal grease discouraged significantly attachment of the ticks onto the piece of cloth. Only 3 ticks were found on the cloth after 5 trials while 41 ticks attached to the untreated cloth. The assay with

Table 5 Toxicity of the tick grease under laboratory conditions

Identity	Percent knocked down ticks Mean ± SD n = 5
Fm 1	20 ± 2
Fm 2	30 ± 3
Fm 3	100 ± 0*
Fm 4	97 ± 3*
Fm 5	-
S	67 ± 3

** Significantly different from standard (s), Amitraz, ANOVA analysis, P < 0.05*

a stimulus showed a similar trend. Only on was found on the treated leg while 46 was observed on the untreated leg.

The tick grease was applied on small scale farmers' cows and results are shown in Tables 8 and 9 to check knockdown effect and protective effect. The results shows that it was able to knockdown all tick types used in the study, *Rhipicephalus, Amblyomma* and *Hyalomma* species. The commercial acaricide failed to knockdown all the ticks Table 8. *Rhipicephalus* and *Amblyomma* showed resistance to the commercial acaricide as these were the ones still remaining on the cows after the period of study. The tick grease also showed good protective effect on non-infested cows Table 9 while the commercial tick grease fails to stop infestation by *Rhipicephalus* species since 6 were found on the cows Table 9. Untreated areas on same cows showed no knockdown and protective effect.

Discussion

Livestock production plays a major role in providing income and food security however it is now under threat because of the effects of ticks (Abbas et al. 2014). Substantial health and economic impacts has been encountered by a number of cattle farmers worldwide (Fouche et al. 2016). Although current synthetic acaricides play a significant role in management of ticks however their wide usage has negative environmental impact and has promoted the development of multi-acaricides resistant ticks. In order to provide an alternative and effective tick controlling tool this work reports development of a herbal tick grease utilizing terpenes and terpenoids as active compounds and

Table 6 Assays without stimulus

Trial	1	2	3	4	5	Mean
Treated cloth	1	0	0	2	0	1 ± 1
Untreated cloth	6	8	10	12	5	8 ± 1

Table 7 Assay with stimulus

Trial	1	2	3	4	5	Mean
Treated leg	0	1	0	0	0	0 ± 0
Untreated leg	6	10	8	10	12	9 ± 2

Table 8 Test on tick infested cows

Treatment	Animal Identity	Description	Initial tick count	After 1 h	After 2 h
Fm3	*Sirika*	Black cow	26	25	All knockdown
	Natha	Black cow white belly	18	10	All knockdown
	Scope	Brown bull	45	45	All knockdown
S	*Sirika*	Black cow	22	10	3
	Natha	Black cow white belly	23	13	6
	Scope	Brown bull	28	10	4
Untreated areas on same cow	*Sirika*	Black cow	17	17	17
	Natha	Black cow white belly	23	21	21
	Scope	Brown bull	14	14	14

S = commercial tick grease bought from local shops

Table 9 Protective effect of the tick grease on susceptible parts (ears, under tail and udder)

Treatment	Animal identity	Description	Day 1	Day 2	Day 5
Fm3	*Sirika*	Black cow	-	-	-
	Natha	Black cow white belly	-	-	-
	Scope	Brown bull	-	-	-
S	*Sirika*	Black cow	-	2	2
	Natha	Black cow white belly	-	-	1
	Scope	Brown bull	1	2	3
Untreated areas on same cow	*Sirika*	Black cow	6	16	16
	Natha	Black cow white belly	3	12	12
	Scope	Brown bull	0	6	10

S = commercial tick grease bought from local shops

its acaricidal activity. Terpenes and terpenoids have been reported to exhibit acaricidal activity in previous studies. Cardosoa et al. (2020) reported inhibition of AChE from resistant and sensitive strains of *R. (B.) microplus* by terpenes such as p-cymene, thymol, carvacrol, and citral. Since tick resistance has been reported for most known commercial acaricides (Abbas et al. 2014) in use currently, there is a need to study formulations with new active compounds.

The active terpenes and terpenoids were bioassay isolated and chemical composition determined using GC–MS. The GC–MS results showed presence of various monoterpenes and sesquiterpenes in the active fractions. *Cissus quadrangularis* exhibited high relative abundances of α-ocimene, copaene, murolene and caryophyllene oxide Table 3. In previous studies essential oil rich in α-ocimene, murolene and caryophyllene exhibited high acaricidal activity against *Oligonychus afrasiaticus* (Gaid et al. 2024). In another study by Branco et al. (2020) an essential oil from *Psidium rufum* with high acaricidal activity against bovine tick *Rhipicephalus microplus* was found to be rich in sesquiterpenes (70.07%), with majority being 1.8-cineole (19.00%); followed by α-longipinene (18.62%).

When the terpenes and terpenoids rich fraction was used to formulate a tick grease it was found that the effective concentration exhibiting greater acaricidal activity was 50% with a percent knockdown effect of 100% under laboratory conditions. While in previous studies acaricidal activity of essential oils were only tested in their natural form under laboratory studies using adult immersion or repellent test (Abdullah et al. 2012; Krengel et al. 2023), the present study formulated a tick grease and tested acaricide activity under both laboratory and field studies to check the feasibility of applicability of the laboratory results. Under field studies the tick grease consisting of 50% terpenes extracts showed good repellence and knockdown effect. The tick grease knocked down and repelled all the ticks including those which were resistant to the commercial acaricide, *Rhipicephalus appendiculatus, Amblyomma variegatum*. During the 5 day period no contact irritation was observed showing that the grease was safe for topical applications. Results of the untreated areas on same cow showed that the terpene and terpenoid tick grease has good efficacy. Monoterpenes and sesquiterpenes arrest tick resistance by a mechanism that is thought to be related to tyramine receptor inhibition (Gross et al. 2017). On testing the protective effect of the tick grease on non-infested cows the tick grease had excellent protective effect for the 5 days chosen for the study showing that chosen base ingredients stabilised the active terpenes very well.

Conclusion

The tick grease consisting of 50% *Cissus quadrangularis* monoterpenes and sesquiterpenes with α-ocimene, copaene, murolene and caryophyllene oxide having high relative abundances exhibited significant acaricidal activity. The tick grease repelled and knocked all ticks including those resistant to the commercial acaricide. The tick grease also showed good protective effect on non-tick infested cows for 5 days. This shows the potential of utilising the monoterpenes and sesquiterpenes from *Cissus quadrangularis* in topical applications to manage tick infestations.

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Author contributions Eve R. Kuvawoga conceptualise the work. Eve R. Kuvawoga and Pamhidzai Dzomba carried out laboratory experiments. Marshall Dhliwayo and Clement Mangwiro managed literature. All the authors carried out semi field studies. Eve R. Kuvawoga, Pamhidzai Dzomba and Clement Mangwiro carried out field studies. Eve R. Kuvawoga and Marshall Dhliwayo wrote the first draft. Pamhidzai Dzomba wrote the second draft. All authors edited and approved the final draft.

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Data availability Most of the data has been provided in this publication. However, more data is available from the corresponding author upon reasonable request.

Declarations

Ethical considerations Ethical approval was obtained from University ethics committee.

Conflict of interest The authors declare no conflicts of interest.

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